Department of Computer Science and Engineering
UIET, CSJM University, Kanpur

Assignment–1: First Unix Laboratory

Preparing Our Computers for C Programming

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Quick Revision

Commands Learned in Class

Command	Purpose	Command	Purpose
`date`	Show the current date and time.	`pwd`	Show the current working directory.
`ls`	Show files and directories.	`cd`	Change the current directory.
`mkdir`	Create a directory.	`touch`	Create an empty file.
`cp`	Copy a file. The original remains.	`mv`	Move or rename a file or directory.

Important Locations and Symbols

`/`	Root of the complete filesystem.	`~`	Your home directory.
`.`	The current directory.	`..`	The parent directory.

Remember: Use `pwd` to check where you are, use `ls` to check what is present, and read every command before pressing Enter.

Purpose of This Lab

You are beginning your studies at UIET, CSJM University, Kanpur. In the coming classes, you will learn to write programs in C.

Before writing programs, you must learn how to create, locate and organize files and directories in your Unix account.

Think of the filesystem as an almirah:

- a **directory** is like a shelf;
- a **subdirectory** is like a box kept on a shelf; and
- a **file** is like a document kept inside a box.

In this lab, you will prepare an organized workspace for your future C programs.

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Problem 1: Your First Day in the UIET Computer Lab

Story

You have logged in to a Unix computer in the UIET laboratory. First, find today's date, your current location and the important places in the filesystem.

Perform the following tasks:

1. Display the current date and time.
2. Display your present working directory.
3. Move to the root directory `/`.
4. Display the contents of the root directory.
5. Display your current location.
6. Return to your home directory using `~`.
7. Display your current location again.
8. Create a directory named **`UIET`**.
9. Use **`ls`** to check that **`UIET`** was created.
10. Use **`ls .`** and observe whether your current location changes.

Think and answer:

- What is the difference between the root directory and your home directory?
- What do `.` and `..` represent?

Problem 2: Prepare Shelves for C Programming

Story

C programming will begin soon. Prepare separate directories for lectures, laboratory work, assignments and projects so that your future programs remain organized.

Inside **`UIET`**, create the following structure:

```
UIET/
‘-- C_Programming/
|-- Lectures/
|   ‘-- introduction.txt
|-- Lab_Work/
|   ‘-- Lab_01/
|       ‘-- hello.c
|-- Assignments/
‘-- Projects/
```

Complete these steps:

1. Enter the **`UIET`** directory.
2. Create **`C_Programming`**.
3. Inside it, create **`Lectures`**, **`Lab_Work`**, **`Assignments`** and **`Projects`**.
4. Inside **`Lectures`**, create **`introduction.txt`**.
5. Return to **`C_Programming`** using `...`
6. Inside **`Lab_Work`**, create **`Lab_01`**.
7. Inside **`Lab_01`**, create **`hello.c`**.
8. Display the complete path of **`Lab_01`**.
9. Return to **`C_Programming`** using `...`

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10. Use `ls` to verify the directories and files.

Think and answer:

- Which command created the empty file `hello.c`?
- What is the complete path of `hello.c`?

Problem 3: Organize Your First C Project

Story

In future classes, you may write C programs to calculate student marks, attendance and canteen bills. Today, prepare and organize the files for this future project.

Inside `Projects`, create a directory named `Campus_Utility`. Prepare the following final structure:

```
Campus_Utility/
|-- Programs/
|   |-- marks.c
|   |-- attendance.c
|   `-- canteen.c
|-- Backup/
|   |-- marks_backup.c
|   `-- canteen_backup.c
`-- Documentation/
|-- project.txt
`-- programming_steps.txt
```

Complete these steps:

1. Inside `Campus_Utility`, create `Programs`, `Backup` and `Notes`.
2. Inside `Programs`, create `marks.c`, `attendance.c` and `canteen.c`.
3. Inside `Notes`, create `project.txt` and `steps.txt`.
4. Copy `marks.c` into `Backup`.
5. Rename its copied version to `marks_backup.c`.
6. Copy `canteen.c` into `Backup`.
7. Rename its copied version to `canteen_backup.c`.
8. Rename `steps.txt` to `programming_steps.txt`.
9. Rename the `Notes` directory to `Documentation`.
10. Use `pwd` and `ls` to verify the final project.

Think and answer:

- What is the difference between `cp` and `mv`?
- Did copying `marks.c` remove its original version?
- Why is keeping a backup of a C program useful?

Final Check

Before completing the laboratory, verify that:

- the `C_Programming` directory is inside `UIET`;
- `hello.c` is inside `Lab_01`;
- the three project files are inside `Programs`;
- the two copied files are inside `Backup`;
- `Notes` has been renamed to `Documentation`.

A well-organized directory is like a well-organized almirah.
When C programming begins, you should always know where your program is stored.

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